

Jueun Song

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Undergraduate student in Computer Science and Engineering with interests in AI, biometric data analysis, information theory, and human-centered computing.

Education

Seoul National University | B.S. Candidate, Computer Science and Engineering

Minor in Biological Sciences; Interdisciplinary Program in Brain-Mind-Behavior

Relevant coursework: Discrete Mathematics, Programming Practice, Computer Programming, Data Structures, Logic Design, Biology I-II, Biology Laboratory I-II

Technical Skills

Programming Languages: C++, Java, Python

Projects

Deep Learning-Based Adaptive Fingerprint Recognition System for Older Adults | AI model design and data analysis

- Designed an age-aware biometric recognition pipeline for older adults with degraded fingerprint patterns.
- Preprocessed public fingerprint datasets with image augmentation to address data scarcity and class imbalance.
- Built a CNN-based age-group classifier and adaptive reconstruction step to improve age-specific fingerprint restoration before recognition.

Wordle Solver Using Shannon Entropy and LLM-Based Feedback Parsing | Search optimization and LLM pipeline tuning

- Implemented a Shannon entropy-based search algorithm with dynamic one-step and two-step pruning.
- Added edge-case heuristics, including sacrificial guesses and small-candidate optimization, to improve search efficiency.
- Optimized an LLM feedback parsing pipeline with Mistral-7B and regex extraction, reducing inference time from 5.95 seconds to 1.56 seconds and averaging 4.024 guesses per game.

Experience & Activities

Volunteer Member | GIV, Seoul National University | 2025-Present

- Participate in international volunteer club activities, including baking service, workshop-based volunteering, and physical activity support for children with developmental disabilities.

Awards

- Presidential Science Scholarship